

Optimality conditions

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Why should I bother to learn this stuff?

- Optimality conditions enable to solve exactly some easy optimization problems (e.g. in microeconomics, some mechanical problems...)
- Optimality conditions are used to derive algorithms for complex problems
- KKT is a certificate of optimality (or non optimality) used by optimization solvers (more than a solving method per se).
- $\implies$ fundamental both for studying optimization as well as other science

Contents

- 1 Optimization problem [BV 4.1]
- 2 Unconstrained case [BV 4.2]
- 3 First order optimality conditions [B.V 5.5]
- 4 Wrap-up

Optimization problem: vocabulary



Generically speaking, an optimization problem is

$$\underset{x \in X}{\text{Min}} \quad f(x) \quad (P)$$

where

- $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is the **objective function** (a.k.a. **cost function**),
- X is the **feasible set**,
- $x \in X$ is an **admissible decision variable** or a **solution**,
- $x^\# \in X$ such that $\text{val}(P) = f(x^\#) = \inf_{x \in X} f(x)$ is an **optimal solution**,
- if $X = \mathbb{R}^n$ the problem is **unconstrained**,
- if X and f are convex, then the problem is **convex**,
- if X is a polyhedron and f linear then the problem is **linear**,
- if X is a convex cone (optionally intersected with affine space) and f linear then the problem is **conic**.

Optimization problem: explicit formulation



The previous optimization problem is often defined explicitly in the following **standard form**

$$\begin{array}{ll} \text{Min}_{\mathbf{x} \in \mathbb{R}^n} & f(\mathbf{x}) \\ \text{s.t.} & g_i(\mathbf{x}) = 0 \quad \forall i \in [n_E] \\ & h_j(\mathbf{x}) \leq 0 \quad \forall j \in [n_I] \end{array} \quad (P)$$

with

$$X := \{\mathbf{x} \in \mathbb{R}^n \mid \forall i \in [n_E], \quad g_i(\mathbf{x}) = 0, \quad \forall j \in [n_I], \quad h_j(\mathbf{x}) \leq 0\}.$$

- (P) is a **differentiable optimization problem** if f and $\{g_i\}_{i \in [n_E]}$ and $\{h_j\}_{j \in [n_I]}$ are differentiable.
- (P) is a **convex differentiable optimization problem** if f , and h_j (for $j \in [n_I]$) are convex differentiable and g_i (for $i \in [n_E]$) are affine.
- ♣ Exercise: Show that in this case X is convex.

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- We can always write an abstract optimization problem in standard form (exercise!)
- For a given optimization problem there is an infinite number of possible standard forms (exercise!)
- We can always find an equivalent problem in dimension $\mathbb{R}^{n+1}$ with linear cost (exercise!)
- A minimization problem with $X = \emptyset$ has value $+\infty$ (by convention)
- A minimization problem has value $-\infty$ iff there exists a sequence $x_n \in X$ such that $f(x_n) \rightarrow -\infty$
- Maximizing f is just minimizing $-f$

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Theorem

Assume that $f : \mathbb{R}^n \rightarrow \bar{\mathbb{R}}$ is differentiable at $x^\#$.

- 1 If $x^\#$ is an unconstrained local minimizer of f then $\nabla f(x^\#) = 0$.
- 2 If in addition f is convex, then $\nabla f(x^\#) = 0$ iff $x^\#$ is a global minimizer.

Proof:

- 1 Assume $\nabla f(x^\#) \neq 0$. DL of order 1 at $x^\#$ show that $f(x^\# - t\nabla f(x^\#)) < f(x^\#)$ for $t > 0$ small enough.
- 2 $f(y) \geq f(x^\#) + \langle \nabla f(x^\#), y - x^\# \rangle$.



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Then $x^\#$ is a minimizer of f on X iff there exists $g \in \partial f(x^\#)$ such that $-g \in N_X(x^\#)$.



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proof : The technical assumption ensures that $\partial(f + \mathbb{I}_X) = \partial f + \partial(\mathbb{I}_X)$.

As $\partial(\mathbb{I}_X) = N_X$, we have, $0 \in \partial(f + \mathbb{I}_X)(x^\#)$ iff there exists $g \in \partial f(x^\#)$ such that $-g \in N_X(x^\#)$.

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From local minimality to a tangent-direction condition



Consider

$$(P) \quad \min_{x \in X} f(x),$$

where $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is differentiable and $X \subset \mathbb{R}^n$ is closed. Let $x \in X$ be a **local minimizer**.

Take any sequence of unit vectors $d_k \rightarrow d$ such that

$$x + \frac{1}{k}d_k \in X \quad \text{for } k \text{ large enough.}$$

Then $x + \frac{1}{k}d_k \rightarrow x$, hence by local minimality,

$$f(x + \frac{1}{k}d_k) \geq f(x) \quad \text{for } k \text{ large enough.}$$

Since f is differentiable at x ,

$$f(x + \frac{1}{k}d_k) = f(x) + \frac{1}{k}\langle \nabla f(x), d_k \rangle + o(\frac{1}{k}).$$

Letting $k \rightarrow \infty$ (so $k o(1/k) \rightarrow 0$ and $d_k \rightarrow d$) yields

$$\langle \nabla f(x), d \rangle \geq 0.$$

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Tangent cones



For $f : \mathbb{R}^n \rightarrow \mathbb{R}$, we consider an optimization problem of the form

$$\underset{x \in X}{\text{Min}} \quad f(x).$$

Definition (Bouligand cone)

We say that $d \in \mathbb{R}^n$ is **tangent** to X at $x \in X$ if there exists a sequence $x_k \in X$ converging to x and a sequence $t_k \searrow 0$ such that

$$d = \lim_k \frac{x_k - x}{t_k}.$$

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Let $T_X(x)$ be the **tangent cone** of X at x , that is, the set of all tangent vectors to X at x .

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Let $T_X(x)$ be the **tangent cone** of X at x , that is, the set of all tangent vectors to X at x .

Equivalently,

$$T_X(x) = \{ d \in \mathbb{R}^n \mid \exists t_k \searrow 0, \exists d_k \rightarrow d, x + t_k d_k \in X \}$$

Optimality conditions - differentiable case

Consider a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and the optimization problem

$$(P) \quad \underset{x \in X}{\text{Min}} \quad f(x).$$

If $x^\# \notin \text{int}(X)$ we do not necessarily need to have $\nabla f(x^\#) = 0$, indeed we just to have $\langle d, \nabla f(x^\#) \rangle \geq 0$ for all "admissible" direction d .

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Theorem

Assume that f is differentiable at $x^\#$.

① If $x^\#$ is a local minimizer of (P) we have

$$\nabla f(x^\#) \in [T_X(x^\#)]^\oplus. \quad (*)$$

② If f and X are both convex, and $(*)$ holds, then $x^\#$ is an optimal solution of (P)

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♠ Exercise: Prove this result.



Let $K_X^{ad}(x)$ be the cone of **admissible** direction

$$K_X^{ad}(x) := \{ t(y - x) \in \mathbb{R}^n \mid y \in X, \quad t \geq 0 \}$$

Lemma

If $X \subset \mathbb{R}^n$ is convex, and $x \in X$, we have

$$T_X(x) = \overline{K_X^{ad}(x)}.$$

Recall that

$$T_X(x) = \{ d \in \mathbb{R}^n \mid \exists t_k \searrow 0, \exists d_k \rightarrow d, x + t_k d_k \in X \}$$

♠ Exercise: Prove this lemma

Differentiable constraints



We consider the following set of admissible solution

$$X = \left\{ \mathbf{x} \in \mathbb{R}^n \mid g_i(\mathbf{x}) = 0, i \in [n_E] \quad h_j(\mathbf{x}) \leq 0, j \in [n_I] \right\},$$

where g and h are differentiable functions.

Recall that the tangent cone is given by

$$T_X(\mathbf{x}) = \left\{ \mathbf{d} \in \mathbb{R}^n \mid \exists t_k \searrow 0, \exists \mathbf{d}_k \rightarrow \mathbf{d}, g(\mathbf{x} + t_k \mathbf{d}_k) = 0, h(\mathbf{x} + t_k \mathbf{d}_k) \leq 0 \right\}$$

We define the **linearized tangent cone**

$$T_X^\ell(\mathbf{x}) := \left\{ \mathbf{d} \in \mathbb{R}^n \mid \begin{aligned} \langle \nabla g_i(\mathbf{x}), \mathbf{d} \rangle &= 0, \forall i \in [n_E] \\ \langle \nabla h_j(\mathbf{x}), \mathbf{d} \rangle &\leq 0, \forall j \in I_0(\mathbf{x}) \end{aligned} \right\}$$

where

$$I_0(\mathbf{x}) := \left\{ j \in [n_I] \mid h_j(\mathbf{x}) = 0 \right\}.$$



We always have

$$T_X(x) \subset T_X^\ell(x).$$

♣ Exercise: Prove it.

We say that the constraints are qualified¹ at x if

$$T_X(x) = T_X^\ell(x).$$

Constraint qualifications ensure the existence of Lagrange multipliers at local minima.

¹This is Abadie CQ, there is a weaker version called Guignard CQ: $(T_X(x))^\oplus = (T_X^\ell(x))^\oplus$.



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Recall that g and h are assumed differentiable.

We denote the index set of **active constraints** at x

$$I_0(x) := \{j \in [n_I] \mid h_j(x) = 0\}.$$

The following conditions are sufficient qualification conditions at x :

- ① g and h_i for $i \in I_0(x)$ are affine in a neighborhood of x ;
- ② (Slater) g is affine, h_j are convex, and there exists x_S such that $g(x_S) = 0$ and $h_j(x_S) < 0$;
- ③ (Mangasarian-Fromowitz) For all $\alpha \in \mathbb{R}^{n_E}$ and $\beta \in \mathbb{R}_{+}^{n_{I_0(x)}}$,

$$\sum_{i \in [n_E]} \alpha_i \nabla g_i(x) + \sum_{j \in I_0(x)} \beta_j \nabla h_j(x) = 0 \quad \implies \quad \alpha = 0 \text{ and } \beta = 0$$

Under constraint qualification, the optimality condition reads

$$\nabla f(x) \in [T_X^\ell(x)]^\oplus$$

where

$$T_X^\ell(x) = \{ d \in \mathbb{R}^n \mid \underbrace{\langle \nabla g_i(x), d \rangle = 0, i \in [n_E] \quad \langle \nabla h_j(x), d \rangle \leq 0, j \in I_0(x)}_{= A_x d \in C} \}.$$

with $A_x = \begin{pmatrix} ((\nabla g_i(x))^\top)_{i \in [n_E]} \\ ((\nabla h_j(x))^\top)_{j \in I_0(x)} \end{pmatrix}$ and $C = \{0\}^{n_E} \times (\mathbb{R}_-)^{n_I}$.

♣ Exercise: Show that $C^\oplus = \mathbb{R}^{n_E} \times (\mathbb{R}_-)^{|I_0(x)|}$

Expliciting the optimality condition



Recall that the positive **dual** cone of a set K is

$$K^{\oplus} := \{ \mathbf{d} \in \mathbb{R}^n \mid \langle \mathbf{d}, \mathbf{x} \rangle \geq 0, \forall \mathbf{x} \in K \}.$$

Let C be a closed convex cone. Consider

$$K = A^{-1}C := \{ \mathbf{x} \in \mathbb{R}^n \mid A\mathbf{x} \in C \},$$

then

$$K^{\oplus} = \{ A^{\top} \boldsymbol{\lambda} \mid \boldsymbol{\lambda} \in C^{\oplus} \}.$$

♣ Exercise: prove it.

Hence,

$$\nabla f(\mathbf{x}) \in \underbrace{[T_{\mathbf{x}}^{\ell}(\mathbf{x})]}_{A_{\mathbf{x}}^{-1}C}^{\oplus}$$

$$\iff \exists \boldsymbol{\lambda} \in C^{\oplus}, \quad \nabla f(\mathbf{x}) + A_{\mathbf{x}}^{\top} \boldsymbol{\lambda} = 0$$

$$\iff \exists \boldsymbol{\lambda} \in \mathbb{R}^{n_E}, \exists \boldsymbol{\mu} \in \mathbb{R}_+^{l_0(\mathbf{x})} \quad \nabla f(\mathbf{x}) + \sum_{i=1}^{n_E} \lambda_i \nabla g_i(\mathbf{x}) + \sum_{j=1}^{l_0(\mathbf{x})} \mu_j \nabla h_j(\mathbf{x}) = 0.$$

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Recall that the positive **dual** cone of a set K is

$$K^{\oplus} := \{ \textcolor{brown}{d} \in \mathbb{R}^n \mid \langle \textcolor{brown}{d}, \textcolor{blue}{x} \rangle \geq 0, \forall \textcolor{blue}{x} \in K \}.$$

Let C be a closed convex cone. Consider

$$K = A^{-1}C := \{ \textcolor{blue}{x} \in \mathbb{R}^n \mid A\textcolor{blue}{x} \in C \},$$

then

$$K^{\oplus} = \{ A^{\top} \textcolor{violet}{\lambda} \mid \textcolor{violet}{\lambda} \in C^{\oplus} \}.$$

♣ Exercise: prove it.

Hence,

$$\nabla f(\textcolor{brown}{x}) \in \underbrace{[T_X^{\ell}(\textcolor{brown}{x})]^{\oplus}}_{A_{\textcolor{brown}{x}}^{-1}C}$$

$$\iff \exists \textcolor{violet}{\lambda} \in C^{\oplus}, \quad \nabla f(\textcolor{brown}{x}) + A_{\textcolor{brown}{x}}^{\top} \textcolor{violet}{\lambda} = 0$$

$$\iff \exists \textcolor{violet}{\lambda} \in \mathbb{R}^{n_E}, \exists \textcolor{violet}{\mu} \in \mathbb{R}_+^{l_0(\textcolor{brown}{x})} \quad \nabla f(\textcolor{brown}{x}) + \sum_{i=1}^{n_E} \textcolor{violet}{\lambda}_i \nabla g_i(\textcolor{brown}{x}) + \sum_{j=1}^{l_0(\textcolor{brown}{x})} \textcolor{violet}{\mu}_j \nabla h_j(\textcolor{brown}{x}) = 0.$$

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$$\iff \exists \textcolor{violet}{\lambda} \in \mathbb{R}^{n_E}, \exists \textcolor{violet}{\mu} \in \mathbb{R}_+^{l_0(\textcolor{brown}{x})} \quad \nabla f(\textcolor{brown}{x}) + \sum_{i=1}^{n_E} \textcolor{violet}{\lambda}_i \nabla g_i(\textcolor{brown}{x}) + \sum_{j=1}^{l_0(\textcolor{brown}{x})} \textcolor{violet}{\mu}_j \nabla h_j(\textcolor{brown}{x}) = 0.$$



Theorem (KKT)

Assume that the objective function f and the constraint function g_i and h_j are differentiable. Assume that the constraints are qualified at $\mathbf{x}$.

Then if $\mathbf{x}$ is a local minimum of

$$\min_{\tilde{\mathbf{x}} \in \mathbb{R}^n} \left\{ f(\tilde{\mathbf{x}}) \mid g_i(\tilde{\mathbf{x}}) = 0, \forall i \in [n_E] \quad h_j(\tilde{\mathbf{x}}) \leq 0, \forall j \in [n_I] \right\}$$

then there exists dual variables λ, μ such that

$$(KKT) \quad \begin{cases} \nabla f(\mathbf{x}) + \sum_{i=1}^{n_E} \lambda_i \nabla g_i(\mathbf{x}) + \sum_{j=1}^{n_I} \mu_j \nabla h_j(\mathbf{x}) = 0 & \nabla_{\mathbf{x}} \mathcal{L} = 0 \\ g(\mathbf{x}) = 0, \quad h(\mathbf{x}) \leq 0 & \text{Primal feasibility} \\ \lambda \in \mathbb{R}^{n_E}, \quad \mu \in \mathbb{R}_+^{n_I} & \text{dual feasibility} \\ \mu_j h_j(\mathbf{x}) = 0 \quad \forall j \in [n_I] & \text{complementarity constraint} \end{cases}$$



Theorem (KKT is necessary and sufficient in convex case)

Assume that f and h_i are **convex** differentiable functions, and that g_j are **affine** functions. Moreover, assume that the **Slater's condition** holds: $\exists x_S \in \mathbb{R}^n$ such that

$$g_i(x_S) = 0 \quad \forall i \in [n_E], \quad h_j(x_S) < 0 \quad \forall j \in [n_I].$$

Then x^* is a (global) minimum of

$$\min_{\tilde{x} \in \mathbb{R}^n} \left\{ f(\tilde{x}) \mid g_i(\tilde{x}) = 0 \quad \forall i \in [n_E], \quad h_j(\tilde{x}) \leq 0 \quad \forall j \in [n_I] \right\}$$

if and only if there exist dual variables λ, μ such that (x^*, λ, μ) satisfy (KKT).

Exercise

Solve the following optimization problem

$$\text{Min}_{x,y \in \mathbb{R}^2} \quad (x-1)^2 + (y-2)^2$$

$$x \leq y$$

$$x + 2y \leq 2$$

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What you have to know

- Basic vocabulary: objective, constraint, admissible solution, differentiable optimization problem
- First order necessary KKT conditions

What you really should know

- What is a tangent cone
- Sufficient qualification conditions (linear and Slater's)
- That KKT conditions are sufficient in the convex case

What you have to be able to do

- Write the KKT condition for a given explicit problem and use them to solve said problem

What you should be able to do

- Check that constraints are qualified